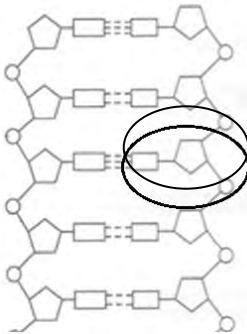
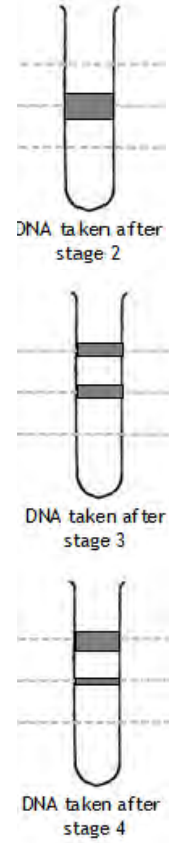


Question number	Answer	Additional guidance	Mark
4(a)(i)	circle drawn around R, the attached sugar and a phosphate group	 ACCEPT the phosphate group on either C3 or C5	(1)

Question number	Answer	Mark			
4(a)(ii)	The only correct answer is D <table border="1" data-bbox="719 997 1767 1120"> <tr> <td>phosphodiester</td><td>covalent</td><td>hydrogen</td></tr> </table> <i>A is incorrect because S is a phosphodiester bond, T is a covalent bond and U is a hydrogen bond</i> <i>B is incorrect because S is a phosphodiester bond and U is a hydrogen bond</i> <i>C is incorrect because T is a covalent bond and U is a hydrogen bond</i>	phosphodiester	covalent	hydrogen	(1)
phosphodiester	covalent	hydrogen			

Question number	Answer	Mark
4(a)(iii)	The only correct answer is C Thymine <i>A is incorrect because adenine is complementary to thymine</i> <i>B is incorrect because adenine is complementary to thymine</i> <i>D is incorrect because adenine is complementary to thymine</i>	(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	A diagram that shows the following points: - a band the same width as stage 1 in the middle of the tube (1) - bands drawn at the top and middle of tube (1) - both bands narrower than stage 1 (1) - bands drawn at the top and middle of tube (1) - top band drawn narrower than stage 1 but wider than stage 3 and lower band drawn narrower than stage 3 (1)	 DNA taken after stage 2 DNA taken after stage 3 DNA taken after stage 4	(5)

Question number	Answer	Mark
4(b)(ii)	The only correct answer is B <i>A is incorrect because neither DNA molecule is made of all heavy nitrogen or light nitrogen</i> <i>C is incorrect because neither DNA molecule is made of all heavy nitrogen or light nitrogen + bands are too wide</i> <i>D is incorrect because it has only one band</i>	(1)